ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

HANDLING NON-CONJUGATE PRIORS

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Binomial with an Arbitrary Prior

Coin of unknown bias θ. Use any prior q_o on $[0, 1]$:

See n observations, with h heads and t tails. Posterior:

$$q_n(\theta) \propto q_o(\theta) \cdot \theta^h \cdot (1 - \theta)^t$$

How to answer questions like "What is $\Pr(\theta < 1/2)$?"

1. Fine gridding of $[0, 1]$, approximate q_o, q_n by discrete distribution over grid points.
2. Most general: Sample from q_n.